TEST DESIGN

1. Equivalence Partitioning (EP)

TCI	Parameter	Equivalence Partition	Expected Behavior
EP1	resourceId	Null	Return false
EP2	resourceId / resource	Non-null UUID; resource unavailable	Return false
EP3	resourceId / resource	Non-null UUID; resource available	Proceed with status update and email
EP4	Status update	Update fails	Throw DatabaseFailureException
EP5	Email sending	Email operation fails	Throw EmailFailureException

1.1 Equivalence Partitioning Test Cases

Test Case	Test Coverage	resourceId	Resource Available	Status Update	Email Sending	Expected Result
TC1	EP1	null	N/A	N/A	N/A	Return false
TC2	EP2	Valid UUID	false	N/A	N/A	Return false
TC3	EP3	Valid UUID	true	Succeeds	Succeeds	Return true
TC4	EP4	Valid UUID	true	Fails	Not attempted	Throw DatabaseFailureException
TC5	EP5	Valid UUID	true	Succeeds	Fails	Throw EmailFailureException

2. Decision Table (DT).

Causes / Effects	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
resourceId == null	T	F	F	F	F
Resource available	–	F	T	T	T
Status update succeeds	–	–	F	T	T
Email sending succeeds	–	–	–	F	T
Result	Return false	Return false	DatabaseFailureException	EmailFailureException	Return true

2.1 Decision Table Test Cases

Test Case	Coverage	resourceId	Resource Available	Status Update Succeeds	Email Succeeds	Expected Result
TC1	Rule 1	null	N/A	N/A	N/A	Return false
TC2	Rule 2	Valid UUID	false	N/A	N/A	Return false
TC3	Rule 3	Valid UUID	true	false	N/A	Throw DatabaseFailureException

TC4	Rule 4	Valid UUID	true	true	false	Throw EmailFailureException
TC5	Rule 5	Valid UUID	true	true	true	Return true

3. Finalized Test Suite

Test Case	Existing Test Method	Coverage	Expected Result
TC1	checkoutResource_nullId_returnsFalse	Null resource ID	Return false; no dependency interactions
TC2	checkoutResource_resourceUnavailable_returnsFalse	Resource unavailable	Return false; no status update or email
TC3	checkoutResource_success_returnsTrue	Successful checkout	Return true; status updated and email sent
TC4	checkoutResource_statusUpdateFails_throwsDatabaseFailureException	Status update fails	Throw DatabaseFailureException; no email
TC5	checkoutResource_emailFails_throwsEmailFailureException	Email returns false	Throw EmailFailureException
TC6	checkoutResource_repositoryThrowsException_propagatesException	Repository throws exception	Propagate RuntimeException; no email
TC7	checkoutResource_emailThrowsException_propagatesException	Email provider throws exception	Propagate EmailFailureException
TC8	checkoutResource_unavailable_doesNotUpdateStatus	No update when unavailable	Status update is never called
TC9	checkoutResource_success_sendsCorrectEmail	Correct recipient and message	Email sent once with expected details
TC10	checkoutResource_statusUpdateFails_doesNotSendEmail	No email after failed update	Throw DatabaseFailureException; no email
TC11	checkoutResource_availabilityCheckThrowsException	Availability check exception	Propagate RuntimeException; no email
TC12	checkoutResource_emailFails_statusWasUpdated	Email failure after update	Throw EmailFailureException; status was updated
TC13	checkoutResource_success_updatesStatusExactlyOnce	Update interaction count	Status updated exactly once